

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461161.8238 +/- 0.009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.037 +/- 0.04
Transit depth (Rp/Rs)^2	0.14 +/- 0.3 [%]
Orbital Inclination (inc)	82.04 +/- 2.88
Airmass coefficient 1 (a1)	24.89 +/- 16.76
Airmass coefficient 2 (a2)	-3.0 +/- 0.86
Scatter in the residuals of the lightcurve fit is	54.73 %
Best Comparison Star	None
Optimal Aperture	7.48
Optimal Annulus	38.39
Transit Duration (day)	0.021 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.037 ± 0.04 vs 0.119 (deviation 1.0σ)
Duration fit vs published	0.021 d vs 0.07 d (deviation 0.77σ)
Mid-time O–C	-9.2 min
Depth SNR	0.5
Residual scatter	54.73 %

Light curve

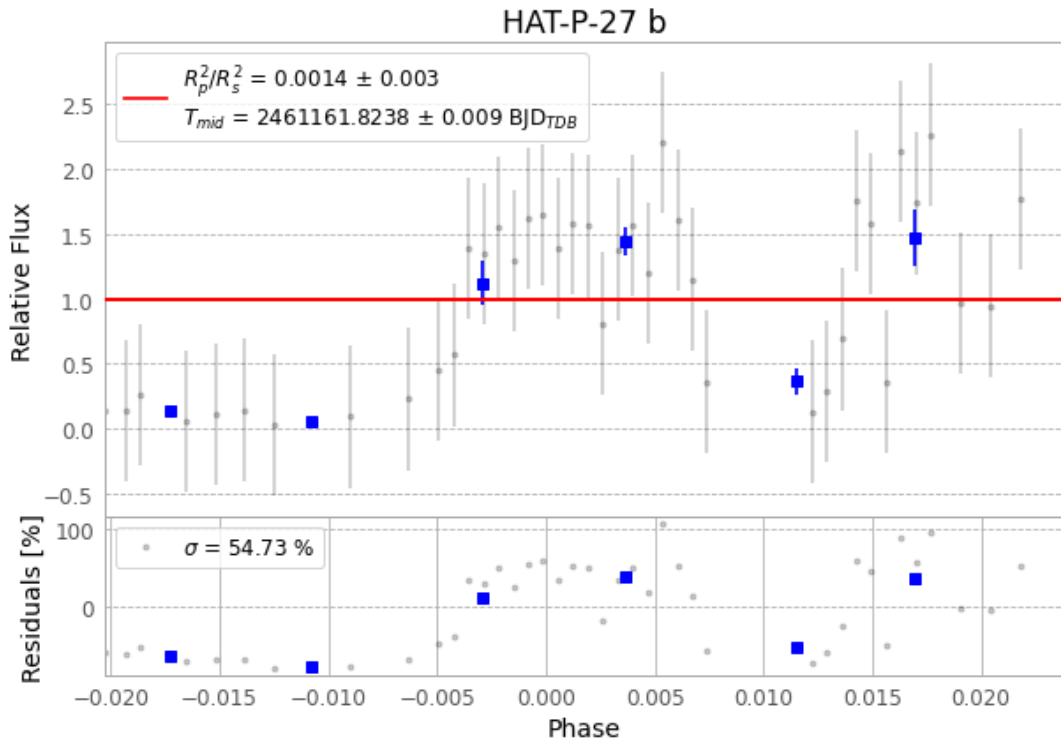


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-05-01.png (SHA-256 15d3f3175eafa452...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [116, 447], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cb761504d8e943c9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

67 FITS frames (44 MB), HATP-27260501060619.FITS → HATP-27260501092115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260501.log (76ab6bcb9d75...), FinalLightCurve_HAT-P-27 b_2026-05-01.png (15d3f3175eaf...), FinalLightCurve_HAT-P-27 b_2026-05-01.csv (ac829049b9fb...)

Compute resources

Wall time 344.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.